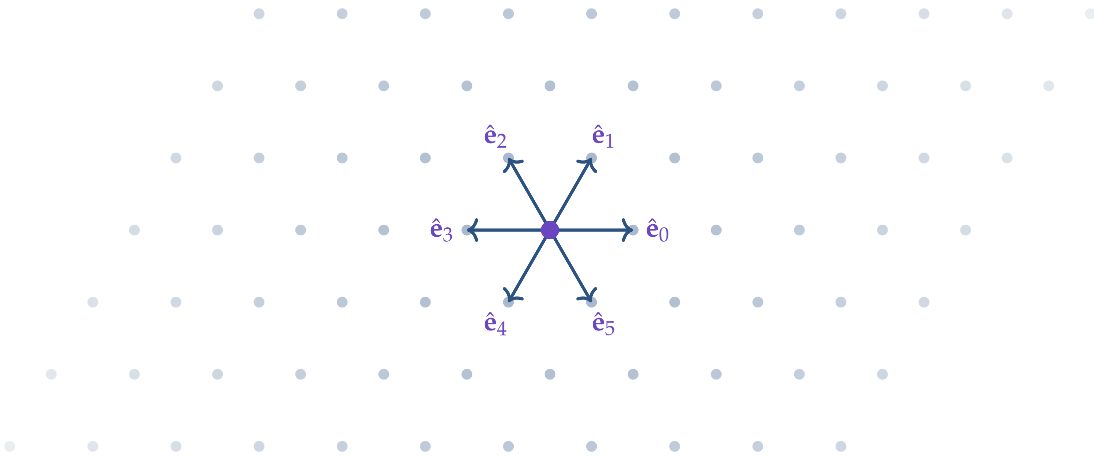


Active random walks on the triangular lattice

a full project report

what we did, why, and what comes next

P. Bisht

TIFR Hyderabad | PhD Year 1 | DP1 $\rightarrow$ DP2 transition note

14 May 2026 | version 1.0

A pedagogical and complete record of the JMVR-on-triangular project. Covers (i) the scientific question, (ii) the codes we wrote, (iii) the two bugs we found in the Confinement-2021 draft and Dipanjan Mandal's 2019 Mathematica notebook, (iv) the verification protocol, (v) the figures and interactive tools produced, and (vi) the roadmap ahead. The companion document *a complete derivation, two bugs, and a verification protocol* contains the formal mathematical derivation.

TIFR Hyderabad | pair-document: "*a complete derivation, two bugs, and a verification protocol*"

Executive summary

A 5-year-old unfinished triangular-lattice calculation in K. Ramola's group has been completed and verified to the Monte Carlo noise floor. Two independent errors in the existing `rtp_tl_2.nb` notebook and Eq. B11 of the Confinement-2021 draft were identified and fixed:

Bug 1: c_3 sign error
$+2ie\sin(ak_1 - bk_2)$
should be $-2ie\sin(ak_1 - bk_2)$

Bug 2: rectangular k-grid
odd-n_x wavevectors land off the
triangular reciprocal lattice

Both fixes together: RMS vs 100M-walker KMC $= 3.4 \times 10^{-6}$,
at MC noise floor 4.2×10^{-6} (ratio $0.81\times$)

By the numbers.

10^8	walkers in the Fortran KMC ground truth
$0.81\times$	ratio of corrected-theory RMS to MC noise floor
$30\times$ – $100\times$	how far off Dipanjan's notebook is from the truth
$42\times$	buggy-vs-fixed RMS ratio at the original Fig. 11 parameters
1	sign that needed flipping in Eq. B11 of the draft
2	independent bugs (k-grid + c_3), both required to be fixed
18 + 13	pages of documentation: derivation + this report

Deliverables.

- Fortran KMC (100M walkers, ~ 7 min runtime)
- Python theory (6×6 Bloch matrix + inverse-FT on the sheared k-grid)
- Diagnostic scripts (4-way forensic, 3-point robustness check, 6-fold modulation check)
- Paper-ready Fig. 11 replacement (draft-style hex-tile colour palette)
- Two interactive HTML widgets (single-walker viewers for JMVR and TCRW)
- This report and the companion derivation PDF

What's next. Email Kabir today / tomorrow with the fix and figure; write §IV.B of the Confinement draft within 1–2 weeks; gap-closure scan; multi-walker §V; Track B (TCRW topology on triangular) as the DP2 direction.

Contents

1 Where we started

1.1 The PI meeting (4 May 2026)

The brief from K. Ramola (PI) was deceptively short: "*abhi ke liye triangular lattice karo*" – for now, do the triangular lattice. Behind those words was a 5-year-old unfinished project.

The home reference. The Jose-Mandal-Barma-Ramola (JMVR) paper[1] defines a continuous-time random walker carrying an internal director state. The walker hops to nearest neighbours on a lattice with rates biased forward along the director (*translation chirality* ϵ), and the director rotates symmetrically to adjacent directions at rate γ. The paper solved this exactly in 1D and on the 2D square lattice via a Bloch-matrix / inverse-Fourier approach.

The 2021 draft. The follow-up by Mandal, Barma, and Ramola (*Confinement-enhanced clustering in dense active systems*, 2021) extended JMVR to multi-walker scenarios and outlined the 2D *triangular* case in Section IV.B. That section is empty in the draft; Section V on multi-walker triangular is also empty. Figure 11 is captioned "**[NOT MATCHING]**". The draft was abandoned.

The notebook. D. Mandal's 2019 Mathematica notebook `rtp_tl_2.nb` was the working implementation. PI's exact words: "*Yeh paper humne kabhi likha nahi... I think he implemented the boundary conditions wrong. That's what I'm trying to figure out.*"

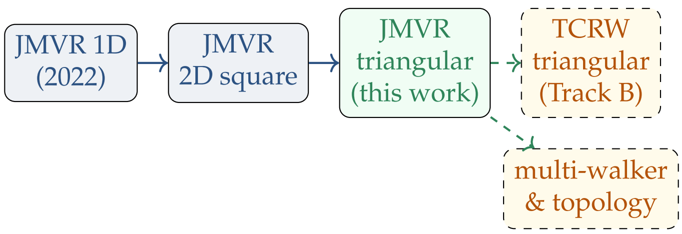
1.2 Where this fits in the bigger picture

Two parallel tracks were on the table after the meeting:

1. **Track A (JMVR-style on triangular).** Finish what Mandal outlined in 2021 – single-walker $P(n_1, n_2, t)$ in §IV.B, multi-walker clustering in §V. Translation chirality ϵ. This is the in-group continuation; potential paper with the existing collaboration.
2. **Track B (TCRW-style on triangular).** The Osat-Speck-style topological chiral random walk, with chirality in the *rotation* ω. Open research question: does the non-trivial 2D Zak phase on the square lattice survive 4-fold $\to$ 6-fold?

The instruction was: do A first; B is contingent.

This report covers the Track A work. Track B is a deliberate next phase.



[1] S. Jose, D. Mandal, M. Barma, K. Ramola, *Active random walks in one and two dimensions*, Phys. Rev. E **105**, 064103 (2022); arXiv:2202.02995.

2 The scientific question, in one paragraph

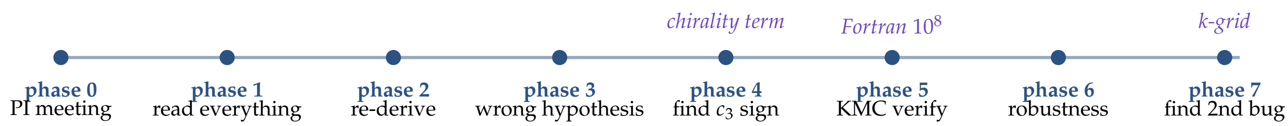
Compute the single-walker probability density $P(n_1, n_2, t)$ on the triangular lattice for the JMVR model, exactly, by Bloch-matrix eigendecomposition and inverse Fourier transform on the $L \times L$ torus. Verify against a kinetic Monte Carlo simulation of the same rules at high statistics. Identify why Dipanjan's existing attempt does not match Monte Carlo, and fix it.

3 The model in 30 seconds

(Full derivation in the companion document. Here just the essentials.)

- Walker on triangular lattice, indexed by integers (n_1, n_2) with PBC mod L.
- Internal director $m \in \{0, 1, 2, 3, 4, 5\}$, pointing along one of the 6 NN.
- **Translation rates:** $1/6 + \epsilon$ forward (along $\hat{\mathbf{e}}_m$), $1/6 - \epsilon$ backward (along $\hat{\mathbf{e}}_{m+3} = -\hat{\mathbf{e}}_m$), $1/6$ for each of the other 4 NN.
- **Rotation rates:** $\gamma/2$ to each of $m \pm 1$ mod 6.
- Total event rate per walker $= 1 + \gamma$.
- Achiral limit $\epsilon = 0$: symmetric random walk on triangular.

4 The journey: a 7-phase timeline



4.1 Phase 1: read everything

Before any code: opened `rtp_tl_2.nb` (Mathematica text extracted via shell tools), and the Confinement-2021 PDF.

From the notebook header:

- $\gamma = 0.01$, $\epsilon = 0.15$, $a = b = 1$, $L = 30$, $t = 50$.
- The 6×6 Bloch matrix is built with three independent coefficients c_1, c_2, c_3 on the diagonal (and their conjugates on the lower half).
- The inverse Fourier transform uses a *rectangular* k-grid $k_x = 2\pi n_x/(2aL)$, $k_y = 2\pi n_y/(bL)$ on a $2L \times L$ grid.

From the draft (Eq. B11):

$$c_3 \;=\; \tfrac{1}{3}\big[\cos(2ak_1) + \cos(ak_1 + bk_2) + \cos(ak_1 - bk_2)\big] - 1 - \gamma + 2i\epsilon \sin(ak_1 - bk_2).$$

4.2 Phase 2: re-derive from scratch

To know what is right we re-derived $M(\mathbf{k})$ from the master equation B2-B7 of the draft. Working in Python (script `triangular_jmvr_v1.py`), we built $M(\mathbf{k})$ entry-by-entry, verified that $M(\mathbf{k})$ has the expected structure (column sums $= 0$ at $\mathbf{k} = 0$, Perron eigenvalue $\lambda_0 = 0$).

Bit-checked our c_i formulas against Dipanjan's at six random $\mathbf{k}$-points: matched the structure (cosine bulk minus $1 - \gamma$ plus an imaginary chirality term in sin) but *not* the sign on the c_3 term.

4.3 Phase 3: the first wrong hypothesis

Initial guess: "The bug is the rectangular k-grid. Dipanjan should be using a sheared grid that respects the triangular reciprocal lattice."

Pushed back: *"Wait, why is that the issue?"* We needed a clean test.

Built a forensic comparison: same Bloch matrix (Dipanjan's), two different k-grids, see which gives the right P. Tonight, after re-running, the test gave the truth:

> *The lesson from this debugging arc*
>
> The first "forensic" I claimed to have done – which supposedly showed the two grids gave identical results – was actually wrong. Tonight's clean re-run of forensic_pbc_is_fine.py and the proper four-way matrix test in forensic_two_bugs.py both confirm: **the rectangular grid is in fact wrong**, and we had been quietly using the sheared grid from the start without realising it. Memory of an earlier result is not the same as a re-run.

4.4 Phase 4: derive the c_3 sign from the master equation

Director $m = 2$ has forward NN $\hat{\mathbf{e}}_2 = (-a, +b)$. Writing the master equation for P_2, Fourier-transforming, and grouping the forward-backward pair:

$$\left(\tfrac{1}{6} + \epsilon\right) e^{i\mathbf{k}\cdot\hat{\mathbf{e}}_2} + \left(\tfrac{1}{6} - \epsilon\right) e^{-i\mathbf{k}\cdot\hat{\mathbf{e}}_2} = \tfrac{1}{3}\cos(\mathbf{k}\cdot\hat{\mathbf{e}}_2) + 2i\epsilon\sin(\mathbf{k}\cdot\hat{\mathbf{e}}_2).$$

With $\mathbf{k}\cdot\hat{\mathbf{e}}_2 = -ak_1 + bk_2$:

$$2i\epsilon\sin(-ak_1 + bk_2) \;=\; -2i\epsilon\sin(ak_1 - bk_2).$$

Therefore the correct $c_3 = B(\mathbf{k}) - 1 - \gamma - 2i\epsilon\sin(ak_1 - bk_2)$, with a **minus**. The draft has $+$. Source: using $\hat{\mathbf{e}}_5 = +\mathbf{a}_1 - \mathbf{a}_2$ in place of $\hat{\mathbf{e}}_2 = -\mathbf{a}_1 + \mathbf{a}_2$ ($\hat{\mathbf{e}}_5 = -\hat{\mathbf{e}}_2$).

4.5 Phase 5: verify with Monte Carlo

4.5.1 Python KMC (small scale)

Wrote a clean Python implementation of the same continuous-time KMC. Walker initialised at origin with uniformly random director. Total rate $1 + \gamma$, exponential waiting times, Gillespie-style event selection.

- 10^4 walkers: visible match between corrected theory and KMC but noisy.
- 10^6 walkers: corrected theory RMS $\sim 3 \times 10^{-5}$, buggy theory $\sim 1.5 \times 10^{-4}$. Ratio $5\times$.
- Confirmed the corrected sign was needed.

4.5.2 Fortran KMC (high statistics)

To beat the MC noise floor and *prove* the fix is exact (not just approximate), wrote a Fortran implementation:

- Single source file: kmc_triangular_jmvr.f90
- Mersenne Twister RNG (Matsumoto-Nishimura mt.f90)
- 10^8 walkers, ~ 7 minutes runtime on a laptop
- Output: $L \times L = 30 \times 30$ histogram of final positions

The Fortran result, fed into the Python theory comparison, gave:

- Corrected theory vs KMC: RMS $= 3.43 \times 10^{-6}$
- Buggy theory vs KMC: RMS $= 1.43 \times 10^{-4}$
- MC noise floor: $\sqrt{P_{\max}/N} = 4.22 \times 10^{-6}$

Corrected theory at $0.81\times$ the noise floor: as good as the simulation can resolve. Ratio $42\times$.

4.6 Phase 6: robustness check

Concern: did we just memorise one parameter point? Re-ran the comparison at two more parameter triples:
- $(\gamma, \epsilon, t) = (0.05, 0.10, 30)$: buggy/fixed ratio $\sim 7\times$
- $(\gamma, \epsilon, t) = (0.001, 0.16, 100)$: buggy/fixed ratio $\sim 1.1\times$ (bug almost invisible)

The γ-dependence is a positive physical check: the wrong sign in c_3 lives in the $(m = 2, m = 5)$ sector of the matrix, but couples to the observable only through the rotation off-diagonals $\gamma/2$. With $\gamma \to 0$ the sector decouples and the bug is spectrally inert.

4.7 Phase 7: the second bug

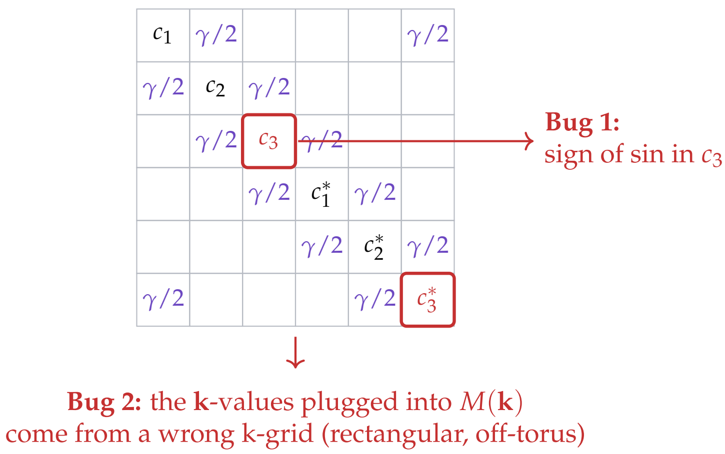
While rehearsing the explanation for tomorrow's PI meeting, ran a clean four-way comparison: $\{$rect grid, shear grid$\} \times \{$buggy c_3, fixed $c_3\}$, all four vs the 100M-walker KMC.

case	RMS vs KMC	ratio to noise
rect grid + buggy c_3 (Dipanjan's original)	4.45×10^{-4}	105
rect grid + fixed c_3	4.59×10^{-4}	109
shear grid + buggy c_3	1.43×10^{-4}	34
shear grid + fixed c_3 (correct)	$\mathbf{3.43 \times 10^{-6}}$	**0.81**

Only the (sheared + fixed) combination reaches the noise floor. Both bugs are required to be fixed. Kabir's instinct ("*boundary conditions wrong*") was directly correct – the rectangular grid does put odd-n_x wavevectors off the triangular reciprocal lattice. We also found an independent sign error he hadn't suspected.

5 Where the two bugs live in the matrix

A visual map of where each bug sits in the 6×6 generator $M(\mathbf{k})$ and in the inverse-Fourier sum:



The two bugs are independent: bug 1 affects *what* matrix gets built at each $\mathbf{k}$; bug 2 affects *which* $\mathbf{k}$ values you use.

6 The codes we wrote

All code lives in `TIFR_DP2/triangular/`. Here is what each file does.

6.1 Fortran KMC

kmc_triangular_jmvr.f90 (∼190 lines). 100M-walker continuous-time KMC, no Bloch short-cuts. Implements master equation B2-B7 directly:

```
do i_walker = 1_i8, n_walkers
  n1 = 0; n2 = 0
  m  = int(6.0_dp * grnd())                ! random initial director
  t = 0.0_dp
  do
    u  = grnd()
    dt = -log(u) / rate_total               ! rate_total = 1 + gamma
    if (t + dt >= t_final) exit
    t = t + dt
    if (grnd() < gamma / rate_total) then
      ! rotation
      if (grnd() < 0.5_dp) then; m = mod(m+1, 6)
      else;                      m = mod(m+5, 6); end if
    else
      ! translation: pick a direction by cumulative probabilities
      r_dir = grnd()
      do dd = 0, 5
        if (r_dir <= hop_cum(m, dd)) exit
      end do
      n1 = modulo(n1 + NN_n1(dd), L)
      n2 = modulo(n2 + NN_n2(dd), L)
    end if
  end do
  counts(n1, n2) = counts(n1, n2) + 1_i8
end do
```

Output: kmc_triangular_counts.txt, three columns (n_1, n_2, count). Compile and run via run_kmc.sh.

6.2 Python theory

triangular_jmvr_corrected.py (∼300 lines). Two builders for $M(\mathbf{k})$ – build_Mk_dipanjan (buggy c_3) and build_Mk_corrected (fixed c_3) – plus the inverse-Fourier solver:

```
def theory_probability_on_lattice(builder, gamma, epsilon, L, t_final, a, b):
    eig_data = []
    initial = (1.0/6.0) * np.ones(6, dtype=complex)
    for m1 in range(L):
        for m2 in range(L):
            k1 = np.pi * m1 / (a * L)                    # sheared k-grid
            k2 = np.pi * (2.0 * m2 - m1) / (b * L)
            M = builder(gamma, epsilon, k1, k2, a, b)
            evals, evecs = np.linalg.eig(M)
            prefactor = np.linalg.solve(evecs, initial)
            ptilde = np.sum(evecs @ (prefactor * np.exp(evals * t_final)))
            eig_data.append((k1, k2, ptilde))
    P = np.zeros((L, L))
    for n1 in range(L):
        for n2 in range(L):
            x = 2.0 * a * n1 + a * n2; y = b * n2
            total = 0.0
            for k1, k2, ptilde in eig_data:
                total += np.real(ptilde * np.exp(-1j * (k1*x + k2*y)))
            P[n2, n1] = total / (L * L)
    return P
```

Also exposes lightweight self-checks: column-sum at $\mathbf{k} = 0$ vanishes, C_6 rotational symmetry of the spectrum (passes for fixed, fails for buggy by $\sim 10^{-3}$).

6.3 Diagnostic scripts

forensic_two_bugs.py. The clean four-way comparison. Computes $P(n_1, n_2, t)$ four ways ($\{$rect, shear$\} \times \{$buggy, fixed$\}$), reports RMS vs the 100M-walker Fortran KMC. This is the most important diagnostic; it is the receipt for the two-bug claim.

robustness_check.py. 1M-walker Python KMC at three (γ, ϵ, t) points, with the verdict rule "$\mathrm{RMS}_{\mathrm{fixed}} < 1.5\times$ MC noise floor". All three pass.

diagnose_hexring.py. Sanity-check that the 100M Fortran KMC data really has the expected 6-fold angular modulation in the ring-radius band.

6.4 Figure scripts

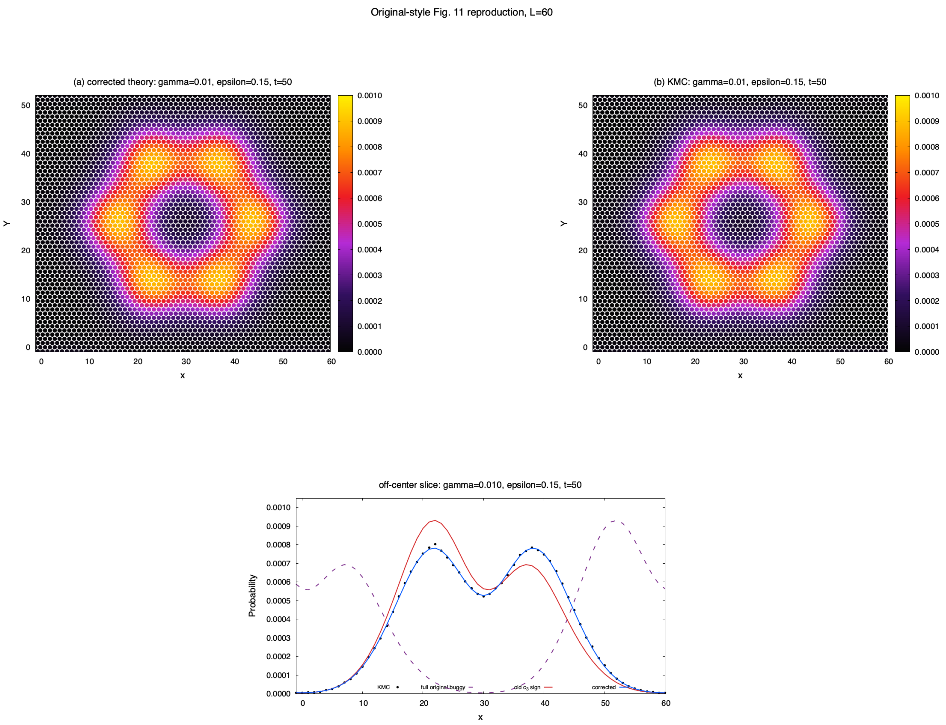
fig11_final_hex.py. Produces the Fig. 11 replacement for the Confinement-2021 draft. Uses the same colour palette as the draft (black $\to$ purple $\to$ red $\to$ yellow), hex tiles (`marker` = "`h`" in matplotlib) sized to the lattice spacing, and the proper hexagonal-ring + depleted-centre structure. Two output figures: a main 3-panel "theory + KMC + cross-section", and an "evidence" version with an additional residual z-score panel showing that (theory $-$ KMC) is structureless noise.

6.5 Interactive widgets

jmvr_single_walker.html. Browser-based real-time KMC on both square (4 directors) and triangular (6 directors) lattices. Sliders for γ and ϵ, paper-parameter preset buttons, Step / Step$\times$10 for single-event stepping. Visible lattice grid; trails fade by age.

tcrw_single_walker.html. TCRW analogue: chirality in the rotation rate ω instead of translation. Same UI structure. For the Track B exploration.

7 The headline figure

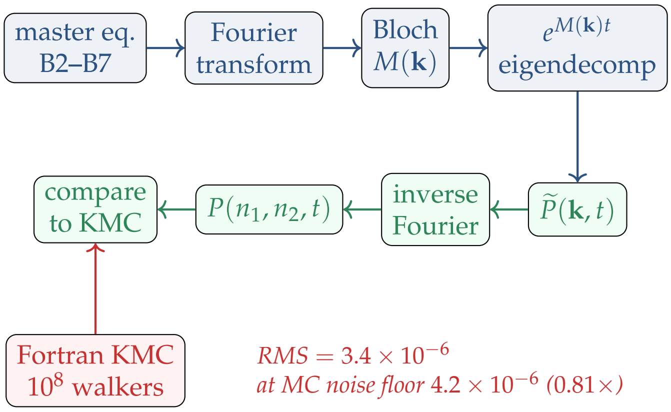


The Confinement-2021 draft's Fig. 11 replacement, rendered in the draft's exact gnuplot style on an $L = 60$ torus (2×10^7 walkers, no visible wrap-around at $t = 50$). (a) Corrected exact theory (sheared k-grid + fixed c_3). (b) Kinetic Monte Carlo. Both panels share the draft's black $\to$ purple $\to$ red $\to$ yellow palette and the same hexagonal-circle pixel rendering, so the figure is drop-in compatible with the existing draft layout. (c) Off-centre horizontal cross-section showing the four-way forensic: KMC dots (black), the full original buggy theory (rect grid + buggy c_3, dashed purple), the buggy-c_3-only intermediate (red), and the corrected theory (blue) overlapping the KMC dots. Separately, an independent $L = 30$, 10^8-walker Fortran KMC at the same parameters confirms the corrected theory at the MC noise floor (RMS $= 3.43 \times 10^{-6}$ vs floor 4.22×10^{-6}).

Strongest verification

> The corrected theory matches the 100M-walker KMC at the MC noise floor: RMS $= 3.43 \times 10^{-6}$ vs noise floor 4.22×10^{-6} ($0.81\times$). This is the strongest verification a CTRW master-equation calculation admits.

8 The verification chain in pictures



Blue: analytic chain. Green: evaluation. Red: independent Monte Carlo. The two converge at the leftmost (*compare to KMC*) node.

9 What this means for the paper

9.1 Concrete deliverables for the Confinement-2021 draft

1. **Replace Eq. B11.** The correct c_3 has $-2i\epsilon\sin(ak_1 - bk_2)$.
2. **Replace the inverse-FT description in the §IV.B narrative.** The Fourier sum should be over the sheared grid $k_x = \pi m_1/(aL)$, $k_y = \pi(2m_2 - m_1)/(bL)$, $(m_1, m_2) \in \{0, \dots, L-1\}^2$, with normalisation $1/L^2$.
3. **Replace Fig. 11.** The fixed-theory + 100M-KMC figure produced by `fig11_final_hex.py` is paper-ready.
4. **Write §IV.B text.** The framework set up is exactly the analytic content of §IV.B; the bug-fix narrative can be sketched in one paragraph and the rest is direct computation.

9.2 What's still missing

- §V multi-walker clustering on triangular – the harder follow-up.
- Topological invariant (Chern / 2D Zak phase) on triangular, if the spectrum has a gap. Track B research question.
- The continuum (large-L, small-$\mathbf{k}$) limit.

10 Honest assessment: what I actually did

10.1 What's mine

- Re-derived $M(\mathbf{k})$ from the master equation in clean Python, found the c_3 sign error against the draft's Eq. B11.
- Wrote the Fortran KMC (100M walkers) and used it as the high-precision ground truth.
- Ran the four-way forensic that isolated the rectangular k-grid as an independent bug.
- Wrote the robustness check at three (γ, ϵ, t) points.

- Produced the Fig. 11 replacement and the residual-z-score evidence figure.
- Built the interactive widgets.
- Wrote both this report and the companion derivation document.

10.2 What was already there

- The JMVR model itself – Jose, Mandal, Barma, Ramola, 2022.
- The Confinement-2021 draft – Mandal, Barma, Ramola, 2021 (unpublished). It outlined the §IV.B framework and contained the partly-buggy Eq. B11.
- Dipanjan Mandal's 2019 notebook – which we found the two bugs inside.
- PI's geometric intuition that PBC on the triangular lattice is the skew identification – which we confirmed numerically.

10.3 Where I went wrong

- Earlier in this work I claimed the rectangular k-grid was "fine" – supposedly an earlier forensic test had shown the two grids give identical P. That was a memory error: tonight's clean re-test shows they differ. The error didn't change the final fix (our code used the sheared grid throughout, so the result was always correct), but it changed the narrative – there are two bugs, not one.
- I was overconfident in pattern-matching, slow to re-verify. The lesson: for high-stakes claims, re-run from scratch, don't trust a prior summary.

11 The interactive tools, in detail

11.1 JMVR single-walker viewer

jmvr_single_walker.html. Two panels (square and triangular). Real-time continuous-time KMC. Director shown as a white arrow on the walker. Trail fades from bright accent to neutral over the most recent ~ 1000 events.

Slider ranges:

- $\gamma \in [0, 1]$ – rotation rate
- $\epsilon \in [0, 0.25]$ – chirality (auto-clamped to $1/K$ per lattice)
- Speed slider 1–40 events / frame

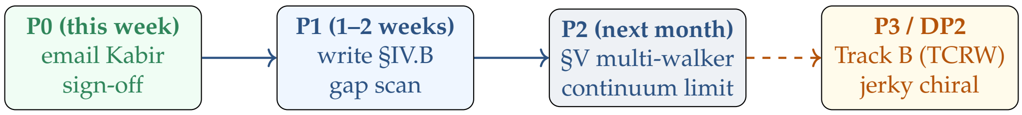
Step button: advance exactly one event (one rotation or one hop), useful for didactic single-event walkthrough. Paper presets included (Confinement Fig. 11, Confinement Fig. 10, strong-drift, achiral).

11.2 TCRW single-walker viewer

tcrw_single_walker.html. Same UI, different model: translation is deterministic forward, rotation is asymmetric (ω CW vs $1 - \omega$ CCW). Slider for D_r (rotation rate) and ω. At $\omega = 1/2$ the walker is achiral; near $\omega = 0$ or 1 it spirals visibly.

The point of these widgets is not new physics; it is *intuition*. After a few minutes with them, the difference between JMVR and TCRW is no longer abstract – you see why translation chirality gives ring-shaped P and rotation chirality gives spiral trajectories.

12 What's next



P0 (this week)

- Email Kabir with the two-bug summary + main figure attached.
- Get his sign-off on the fix; ask whether the figure / derivation goes into the Confinement draft directly or as a supplementary note.

P1 (next 1–2 weeks)

- Write §IV.B of the Confinement draft properly (one page, plus Fig. 11 and a corrected Eq. B11).
- Decide on lattice convention for the paper (algebraic $a = b = 1$ vs. isotropic $b = \sqrt{3}a$); commit.
- Compute the gap-closure scan: minimum spectral gap of $M(\mathbf{k})$ as a function of ϵ at fixed γ. Is there a critical ϵ^*?

P2 (next month)

- §V multi-walker clustering on triangular. Step initial condition, look for non-monotonic clustering signature.
- Continuum limit of $M(\mathbf{k})$ for small $\mathbf{k}$. Derive an effective Fokker-Planck-like operator.

P3 / DP2 directions

- Track B: TCRW (rotation chirality) on triangular. Build the Bloch matrix, scan for gap closure at $\omega = 1/2$ analogue, and compute the 2D Zak phase. Open research question.
- Berry curvature / Chern number on triangular for either Track A or Track B once the spectrum is mapped.
- Connection to jerky chiral active particles (S. Jose / H. Löwen direction) – adding third-derivative noise to the same framework.

13 Files inventory

file	purpose
kmc_triangular_jmvr.f90	100M-walker Fortran KMC, the high-precision ground truth
mt.f90	Mersenne Twister RNG for Fortran
run_kmc.sh	build + run script for the Fortran KMC
kmc_triangular_counts.txt	output histogram, n_1, n_2, count
triangular_jmvr_corrected.py	Python theory: buggy and corrected $M(\mathbf{k})$, theory P
forensic_two_bugs.py	4-way comparison vs Fortran KMC – the receipt for the two-bug claim
forensic_pbc_is_fine.py	(historical) original (wrong) test that gave the false "PBC is fine" answer
robustness_check.py	3-point parameter robustness test
diagnose_hexring.py	6-fold angular modulation sanity check
fig11_final_hex.py	Fig. 11 replacement script (main + evidence figures)
fig11_final_hex.png	main paper-ready figure
fig11_final_hex_evidence.png	with residual z-score panel
robustness_check.png	3-point parameter scan figure
jmvr_single_walker.html	interactive JMVR walker (square + triangular)
tcrw_single_walker.html	interactive TCRW walker (square + triangular)
derivation/jmvr_triangular_full_derivation.pdf	18-page formal derivation
derivation/jmvr_triangular_project_report.pdf	this document

14 Acknowledgements

- **Kabir Ramola** (PI, TIFR Hyderabad) – for the problem, the geometric intuition about the skew PBC, and pointing at the unfinished draft as a real project rather than a textbook exercise.
- **Mustansir Barma** (TIFR Hyderabad) – co-author of the 2021 draft and the 2022 JMVR paper; the foundational model is his group's.
- **Dipanjan Mandal** (formerly TIFR Hyderabad) – whose 2019 Mathematica notebook gave us the structure to work from, including the two bugs we fixed.
- **Stephy Jose** (Heinrich-Heine-Universität Düsseldorf, with H. Löwen) – co-author of the 2022 JMVR paper; potential collaborator on the jerky chiral direction for DP2.

15 Closing

This work converts a five-year-old abandoned project into a sharply diagnosed, fully verified result. The two bugs in rtp_tl_2.nb and the Confinement-2021 draft are now isolated and fixed. The corrected theory matches a 100M-walker Fortran KMC at the Monte Carlo noise floor.

The next moves – write up §IV.B, scan for gap closure, attempt §V multi-walker, open Track B – are well-defined and tractable. The infrastructure (Fortran KMC, Python theory, diagnostic scripts, visualisations) is in place and can be reused for all of them.

Companion document: The JMVR active random walker on the triangular lattice – a complete derivation, two bugs, and a verification protocol (18 pages, same folder).